Spotlight on Technology Case Study

UPS Competes Globally with Information Technology

United Parcel Service (UPS) started out in 1907 in a closet-sized basement office. Jim Casey and Claude Ryan—two teenagers from Seattle with two bicycles and one phone—promised the "best service and lowest rates." UPS has used this formula successfully for more than a century to become the world's largest ground and air package-delivery company. It's a global enterprise with more than 500,000 employees, more than 125,000 delivery vehicles, and 557 aircraft.

Today, UPS delivers close to 25 million packages each day in more than 220 countries and territories. The firm has been able to maintain leadership in small-package delivery services despite stiff competition from FedEx, the US Postal Service, and Amazon by investing heavily in advanced information technology. UPS spends $1 billion annually on IT to maintain a high level of customer service while keeping costs low and streamlining its overall operations.

It all starts with the scannable bar-coded label attached to a package, which contains detailed information about the sender, the destination, and when the package should arrive. Customers can download and print their own labels using special software provided by UPS or by accessing the UPS website or mobile app. Before the package is even picked up, information from the "smart" label is transmitted to one of UPS's computer centers in Mahwah, New Jersey, or Alpharetta, Georgia, and sent to the distribution center nearest its final destination.

The first thing a UPS driver picks up each day is a handheld computer called a Delivery Information Acquisition Device (DIAD), which can access a cell phone network. As soon as drivers log on, their route is downloaded onto the DIAD. The DIAD also automatically captures customers' signatures along with pickup and delivery information. Package tracking information is then transmitted to UPS's computer network for storage and processing. From there, the information can be accessed worldwide to provide proof of delivery to customers or to respond to customer queries. UPS processes an average of 743.5 million tracking requests each day.

Through its automated package tracking system, UPS can monitor and even reroute packages throughout the delivery process. At various points along the route from sender to receiver, barcode devices scan shipping information on the package label and feed data about the progress of the package into the central computer. Customer service representatives are able to check the status of any package from desktop computers. UPS customers can also access this information from the company's website using their own computers or mobile devices.



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UPS created an AI application called DeliveryDefense designed to tackle the growing problem of shipment loss. The application examines characteristics such as location, loss frequency, returns volume, and delivery attempts, grouping them based on similarities. The application predicts the likelihood of a successful delivery, enabling shippers and their customers to reroute at-risk shipments to The UPS Store or a UPS Access Point. The solution can also be built into a customer's ordering platform to improve their delivery outcomes and also helps UPS customers deter package theft. For example, DeliveryDefense helps gold and silver dealer Texas Precious Metals identify addresses that are likely targets for "porch pirates."

An AI-powered app called Network Planning Tools (NPT) helps UPS reroute packages away from bad storms and other trouble spots in its global network. NPT also helps the company to move unusually large shipments efficiently. AI machine learning algorithms handle the massive amounts of data required to move packages around the world and on time, regardless of the weather.

UPS's On-Road Integrated Optimization and Navigation (ORION) is an AI platform for planning and optimizing the routes taken by UPS drivers, based on advanced algorithms, AI machine learning, and deep learning (which is largely self-taught and can recognize patterns and make predictions from very large datasets). UPS vehicles are equipped with systems that capture logistical data. Dispatchers at UPS distribution centers use ORION routing software to create the most efficient delivery route for each driver, taking into account traffic, weather conditions, and the location

of each stop. In a network with 66,000 routes in the United States alone, shaving even one mile off each driver's daily route translates into big savings in time, fuel consumption, miles driven, and carbon emissions. Since ORION's initial deployment, it has saved UPS about 100 million miles and 10 million gallons of fuel per year.

SOURCES: UPS, "Company Facts," About.ups.com, accessed November 18, 2024; Anastasia Morozova, "Chatbots in Logistics and Supply Chain: How Do Companies Like UPS Benefit from Chatbots," Jasoren.com, accessed November 18, 2024; Zach Williamson, "UPS Saving Millions at the Pump, Emphasizes Importance of Planning Ahead," 3newsnow.com, July 18, 2022.

Case Study Questions

1. What are the inputs, processing, and outputs of UPS's package tracking system?
2. What technologies are used by UPS? How are these technologies related to UPS's business strategy? What role is played by AI?
3. What strategic business objectives do UPS's information systems address?
4. What would happen if UPS's information systems were not available?

Let's identify the organization, management, and technology elements in the UPS package tracking system we have just described. The organization element anchors the package tracking system in UPS's sales and production functions (the main product of UPS is a service—package delivery). It specifies the required procedures for identifying packages with both sender and recipient information, taking inventory, tracking the packages en route, and providing package status reports for UPS customers and customer service representatives.

The system must also provide information to satisfy the needs of managers and workers. UPS drivers need to be trained in both package pickup and delivery procedures, as well as how to use the package tracking system so that they can work efficiently and effectively. UPS customers may need some training to use UPS in-house package tracking software, the UPS website, or mobile apps.

UPS's management is responsible for monitoring service levels and costs, and for promoting the company's strategy of combining low cost and superior service. Management decided to use automation to increase the ease of sending a package via UPS and checking its delivery status, thereby reducing delivery costs and increasing sales revenues.

The technology supporting this system consists of handheld computers, bar code scanners, wired and wireless communications networks, desktop computers, UPS's central computer, storage technology for the package delivery data, UPS in-house package tracking software (and other software driving operations such as ORION), and software to access the web. The result is an information system solution to the business challenge of providing a high level of service with low prices in the face of mounting competition.

1.7 Understand the Importance of IoT, Big Data, Cloud Computing, and AI

Among all of these technologies, four developments are especially important today. They are the Internet of Things (IoT), big data, cloud computing, and artificial intelligence (AI). We will take a more detailed look at these in the following section.

The **Internet of Things (IoT)** describes a network of physical objects—"things"—that are embedded with sensors, software, and other technologies enabling them to connect and exchange data with other devices and systems via the Internet. These devices range from ordinary household objects to sophisticated industrial tools. The number of connected IoT devices is projected to reach nearly 22 billion by 2025 (Vailshery, 2023). Digitally linking these devices gives them enough intelligence to make simple decisions and remember particular patterns and routines to be carried out without any human involvement. IoT data is also a major source of data for systems with more sophisticated functionality such as forecasts, predictions, and optimizations.

These connected IoT devices, as well as data from websites and other sources, generate **big data**: large quantities of data (typically characterized by their volume, variety, and the velocity at which they are collected) that can provide useful information but that are often too vast and complex to capture, store, and process using traditional methods.

Internet of Things (IoT)

Network of physical objects (things) embedded with sensors, software, and other technologies enabling them to connect and exchange data with other devices and systems via the Internet.

Big data

Data sets typically characterized by the 3Vs (the extreme volume of data, the wide variety of data, and the velocity at which data must be processed) and which are beyond the ability of a typical DBMS to capture, store, and analyze.